

Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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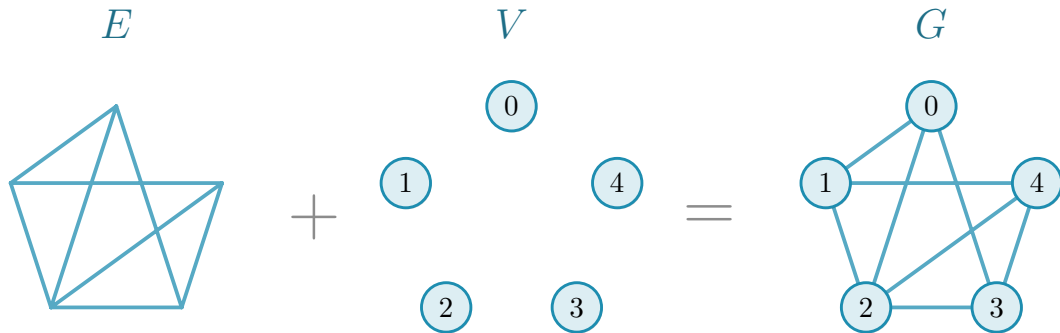
Maximising edge density in graphs
under connectivity constraints



translation

networks with **many links** and **low connectivity**

Graph

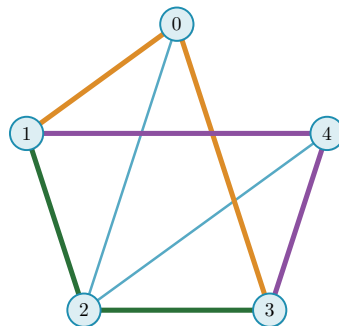


Erdős Problem 915

$$\ell_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \lambda_G^{\max} \leq m-1 \right\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v)$$

$\lambda_G(u, v)$ = the largest number of u - v routes that share **no edge**: one colour each

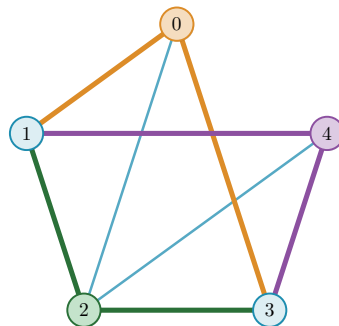


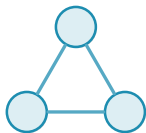
The same question, vertex by vertex

$$k_m(n) = \max_{|V(G)|=n} \{ |E(G)| : \kappa_G^{\max} \leq m-1 \}$$

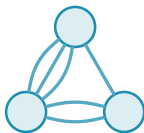
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v)$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**

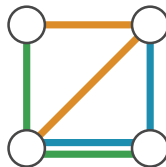




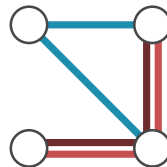
simple graph



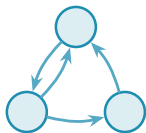
multigraph



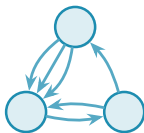
hypergraph



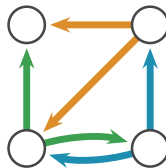
multihypergraph



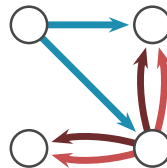
simple digraph



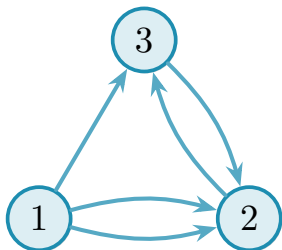
directed multigraph



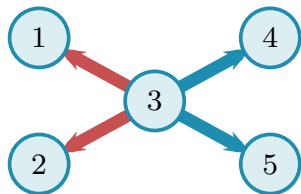
directed hypergraph



directed multihypergraph



$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$



list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

$$\begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

INPUT

||

directed multigraph

max flow
over every pair

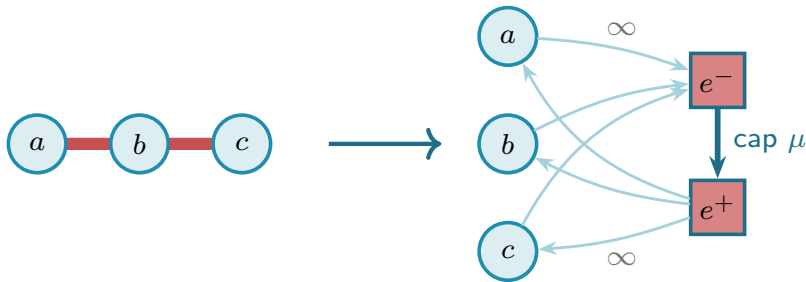
$$\lambda_G^{\max} = 3$$

OUTPUT

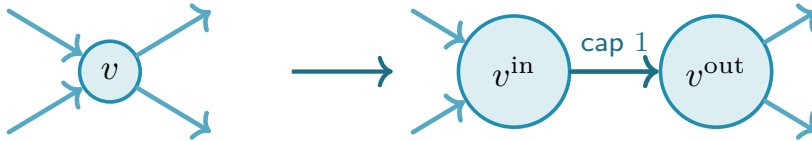
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph



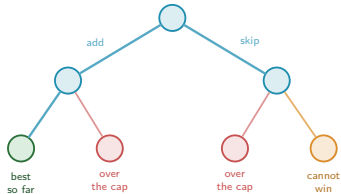
$$\kappa \rightarrow \lambda$$



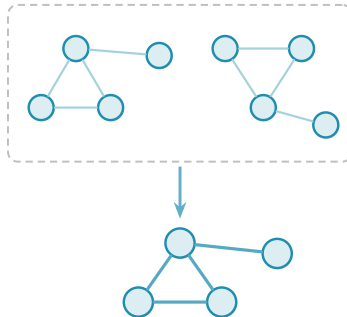
every intermediate vertex is split this way
the flow itself runs from s^{out} to t^{in}

Efficiency: three ways to make the sweep finish

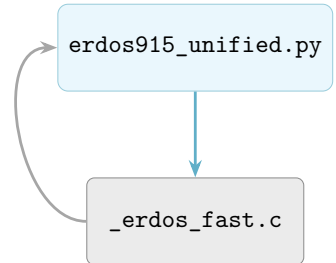
pruning



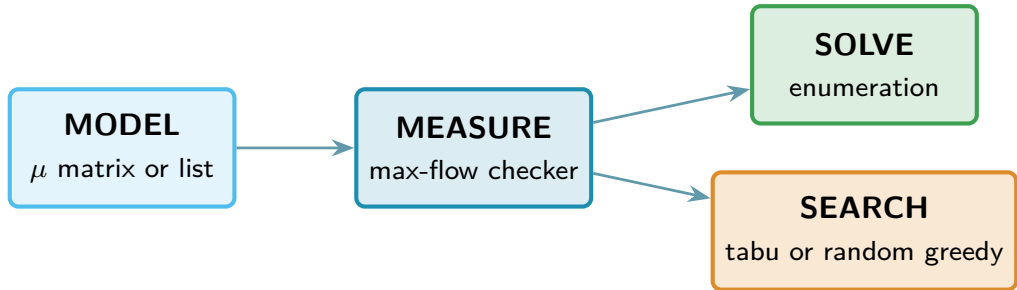
generating search



C: faster max flow

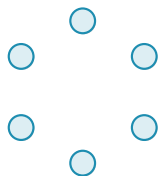


Solving versus searching

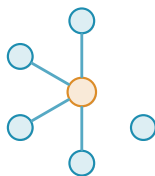


The search: one number per graph

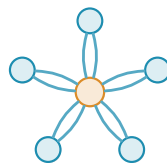
$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$



$$\Phi = 0$$



$$\Phi = -4$$



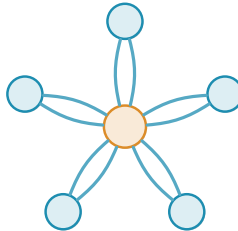
$$\Phi = -10 \text{ at } m = 3$$

Extremal families the search rediscovers



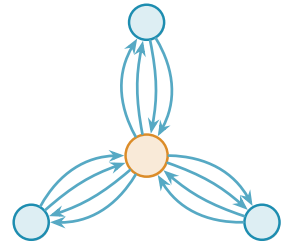
tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

$$n = 6, m = 3, L_3(n) = 10$$

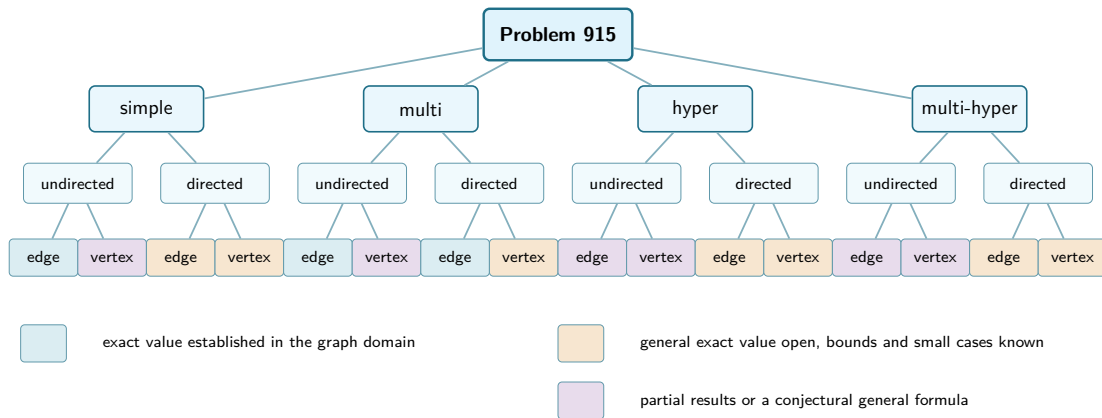


double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

examples of the families, not the saved search witnesses

Status of the sixteen variants



Thank you

`github.com/louisevandenbruwaene/thesis`

Backup: the sixteen variants

HOW TO READ THE NEXT SIXTEEN FRAMES

Each frame takes one variant, fixes a small size, and evaluates the formula there, so the general rule and a number you can check sit side by side.

UPPER the formula, with any condition it needs

LOWER what the construction achieves, as a formula too

1. 2. 3. a sketch of the argument, upper bound first, ending with what attains it

USES the results the sketch leans on, numbered as in the thesis

EXTREMAL the drawing is that witness, and beside it the size and the value both bounds meet at, so every vertex and every edge on it can be counted against that number

Simple graph · edge

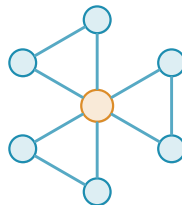
$$\ell_m(n)$$

UPPER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED** $n \geq m$

LOWER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED**

1. a Gomory–Hu tree summarises every pairwise connectivity in $n - 1$ cuts
2. each cut has capacity at most $m - 1$, so together they carry $(m - 1)(n - 1)$
3. every edge is charged to two of those cuts, which halves the count
4. a star with a degree- $(m - 2)$ spoke graph on its leaves attains it

USES Mader 1.2 · Gomory–Hu A.2 · double count A.5–A.7 · hub-and-spokes A.8 · near-regular A.9



$n = 7, m = 3$ **9** **EXTREMAL**

Simple graph · vertex

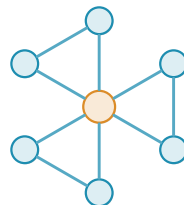
$$k_m(n)$$

UPPER $k_m(n) = \ell_m(n)$ **PROVED** $m \leq 4$

LOWER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED**

1. the edge bound does not transfer: $\kappa \leq \lambda$ makes vertex feasibility the weaker restriction, so there are more graphs to beat
2. for $m \leq 4$ the two values are equal anyway, which is the cited result
3. at $m = 5$ they part: $\left\lfloor \frac{8n}{3} \right\rfloor - 4$ takes over from $n \geq 14$
4. the same drawing attains it, every leaf having degree two

USES Bártfai–Bollobás–Leonard 1.4 · Sørensen–Thomassen 1.5 · degree bound A.3



$n = 7, m = 3$ **9** **EXTREMAL**

Simple digraph · arc

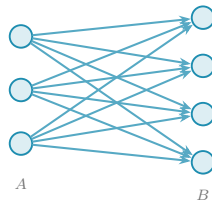
$$\ell_m^{\text{dir}}(n)$$

UPPER $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ **PROVED** $m = 2$

LOWER $\lfloor \frac{n^2}{4} \rfloor$ **PROVED**

1. a reachability skeleton is a subdigraph preserving who reaches whom
2. at $m = 2$ a feasible digraph *is* its own skeleton: drop an arc and the route replacing it is a second disjoint one, breaching the cap
3. a skeleton carries at most $M(n)$ arcs, which bounds the digraph
4. at $n = 7$ both branches of M equal 12, so the bidirected hub and the one-way bipartite digraph each attain it

USES Arc value at $m = 2$ 1.6 · skeleton A.18–A.19 · two-hop A.20 · the two $m = 2$ families A.25 · directed hub A.15 · one-way bipartite A.16



$$n = 7, m = 2 \quad \mathbf{12}$$

EXTREMAL

Simple digraph · vertex

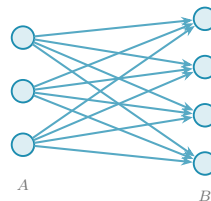
$$k_m^{\text{dir}}(n)$$

UPPER $M(n)$ **PROVED** $m = 2$

LOWER $\left\lfloor \frac{n^2}{4} \right\rfloor$ **PROVED**

1. the routes the skeleton argument produces are internally vertex-disjoint, not merely arc-disjoint
2. so the argument proves the *vertex* bound first, and the arc bound is the consequence rather than the source
3. that direction matters: Whitney would not license carrying a bound the other way
4. the same digraph attains both, so the two values coincide at $m = 2$

USES Vertex value at $m = 2$ 1.7 · sparse skeleton A.19 · deletion monotonicity A.23 · one-way bipartite A.16



$n = 7, m = 2$ **12**

EXTREMAL

Multigraph · edge

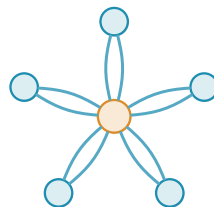
$$L_m(n)$$

UPPER $(m-1)(n-1)$ **PROVED** every n and m

LOWER $(m-1)(n-1)$ **PROVED**

1. the same Gomory–Hu charge: $n-1$ cuts, each of capacity at most $m-1$
2. but parallel copies mean nothing forces an edge to be counted twice
3. so no halving and no floor: the bound is the clean product, and the missing factor of two is exactly what simplicity was buying next door
4. any spanning tree at multiplicity $m-1$ attains it

USES Multigraph edge value 1.3 · Gomory–Hu A.2 · thickened tree A.36



$n = 6, m = 3$ **10**

EXTREMAL

Multigraph · vertex

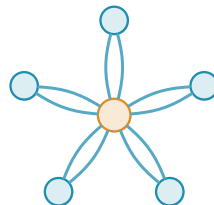
$$K_m(n)$$

UPPER $2(n-1)$ **PROVED** $m=3$

LOWER $(m-1)(n-1)$ **PROVED**

1. between two vertices κ is exactly $\mu(u, v) + \pi(u, v)$: the copies of their own edge, plus the disjoint detours around it
2. feasibility therefore caps those two *together* at $m-1$
3. at $m=3$ that yields the same value as the edge problem
4. a tree edge has no detour, so a doubled tree spends the whole cap on multiplicity and attains it

USES Splitting the vertex measure A.54 · value at $m=2$ A.55 · value at $m=3$ A.57 · incidence rank A.52



$n=6, m=3$ **10**

EXTREMAL

Directed multigraph · arc

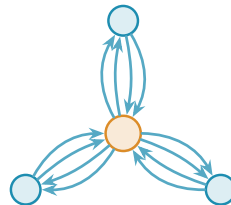
$$L_m^{\text{dir}}(n)$$

UPPER $(m - 1)M(n)$ **PROVED** every n and m

LOWER $(m - 1) \cdot 2(n - 1)$ **PROVED**

1. inside any feasible multigraph there is a sparse reachability skeleton
2. removing one copy of each of its arcs costs every reachable pair one route, so the graph stays feasible one threshold lower
3. $m - 1$ such peels account for every arc, giving $m - 1$ times the $m = 2$ value, and the fractional relaxation is integral so nothing is lost
4. a bidirected tree at multiplicity $m - 1$ attains the linear branch

USES Every n and m 1.13 · sparse skeleton A.19 · deleting a skeleton A.37 · thickened tree A.36 · integrality A.39



$n = 4, m = 3$ **12**

EXTREMAL

Directed multigraph · vertex

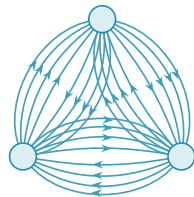
$$K_m^{\text{dir}}(n)$$

UPPER exhaustive enumeration **PROVED**

LOWER $n(n-1)(m+1-n)$ **PROVED**

1. κ splits as $\mu + \pi$ again: direct copies plus disjoint detours
2. with one other vertex there is exactly one detour, so each direction stops at $m-2=4$ rather than at the cap of five
3. the six directions give $6 \times 4 = 24$, and enumeration leaves exactly one isomorphism class
4. it is the complete digraph thickened, the $b=3$ block of A.65

USES Directed incidence value A.64 · splitting the vertex measure A.54 · thickened complete digraph A.65



$$n = 3, m = 6 \quad \mathbf{24}$$

EXTREMAL

Simple hypergraph · edge

$$\ell_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED**

LOWER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** $m-1 \leq \binom{n-2}{r-2}$

1. induction on a single hyperedge cut, with no tree anywhere in it
2. a hyperedge spans r vertices but still serves only one route, so cutting one off divides what remains by $r-1$
3. the reduction to graphs points the wrong way: replacing a hyperedge by a star would let two routes take two of its edges, which Berge forbids
4. the star hypertree attains it, and for $r \geq 3$ simplicity is free

USES Hypergraph edge bound 1.8 · cut induction A.45 · simplicity is free A.46 · sparse hypergraphs A.49 · lower bound A.50



$$n = 9, r = 3, m = 2 \quad 4$$

EXTREMAL

Simple hypergraph · vertex

$$k_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ **PROVED** $m = 2$

LOWER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ **PROVED**

1. at $m = 2$ feasibility says exactly that the incidence graph is a forest, since a cycle in it would already be two disjoint routes
2. a forest on the n vertices together with the $|E|$ hyperedges has at most $n + |E| - 1$ edges
3. each hyperedge contributes r of them, so $r|E| \leq n + |E| - 1$, which rearranges to $|E|(r-1) \leq n-1$
4. the star hypertree's incidence graph is a tree, so it attains it

USES Vertex value at $m = 2$ 1.9 · at $m = 3$ 1.10 · incidence rank A.52 · Menger for multihypergraphs A.44



$n = 9, r = 3, m = 2$ **4**

EXTREMAL

Multihypergraph · edge

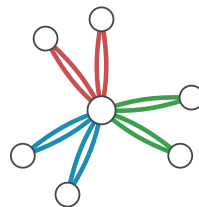
$$L_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** unchanged by repetition

LOWER $\frac{(m-1)(n-1)}{r-1}$ **PROVED** $(r-1) \mid (n-1)$

1. the cut induction never asks whether hyperedges are distinct
2. so the bound is word for word the one next door, and repetition cannot move it
3. what repetition changes is which shapes reach it: the divisibility condition replaces the supply of distinct hyperedges
4. the star hypertree with each hyperedge taken $m-1$ times attains it

USES Hypergraph edge bound 1.8 · cut induction A.45 · multihypergraph lower bound A.51 · twelve copies on six vertices A.48



$$n = 7, r = 3, m = 3 \quad \mathbf{6}$$

EXTREMAL

Multihypergraph · vertex

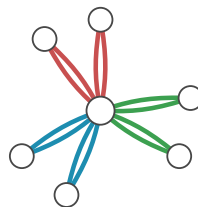
$$K_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ **PROVED** $m = 3$

LOWER $\frac{(m-1)(n-1)}{r-1}$ **PROVED** $(r-1) \mid (n-1)$

1. the incidence graph again, but now its cycle rank is what needs bounding
2. the rank is controlled by decomposing the graph at its cut vertices and at its separating pairs
3. repetition buys nothing at $m = 2$, where two copies of one hyperedge are already two separated routes
4. from $m = 3$ it adds attainment cases, and the repeated star is one

USES Vertex value at $m = 3$ 1.10 · incidence rank A.52 · multihypergraph lower bound A.51



$$n = 7, r = 3, m = 3 \quad \mathbf{6}$$

EXTREMAL

Directed simple hypergraph · arc

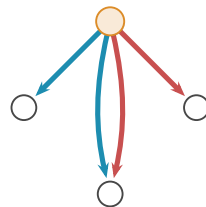
$$\ell_m^{\text{dir}(r)}(n)$$

UPPER exhaustive enumeration **PROVED**

LOWER $n \cdot \min(m-1, \binom{n-1}{r-1})$ **PROVED**

1. every route out of a vertex has to start with one of that vertex's own outgoing hyperedges
2. deleting those cuts it off, so the outgoing count alone caps every pair it reaches, and giving each vertex $\min(m-1, \binom{n-1}{r-1})$ of them is feasible at any size
3. for general n the matching upper bound is only asymptotic: a shadow digraph carries the route cap onto two-step routes
4. at this size enumeration closes the gap, and $4 \times 2 = 8$

USES Directed hypergraph constant 1.14 · first bounds A.67 · bounded outgoing count A.68 · one-step shadow A.71 · two-step budget A.28



$$n = 4, r = 3, m = 3 \quad 8$$

EXTREMAL

one tail of the four, each doing the same

Directed simple hypergraph · vertex

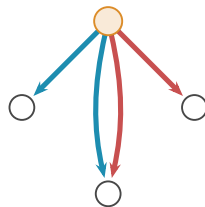
$$k_m^{\text{dir}(r)}(n)$$

UPPER exhaustive enumeration **PROVED**

LOWER $n \cdot \min(m-1, \binom{n-1}{r-1})$ **PROVED**

1. the outgoing cut never looks at interior vertices at all
2. so the very same argument establishes the arc bound and the vertex bound in one go, and the witness needs no adjustment
3. the shadow argument behind the general growth rate re-runs under the stricter separation for the same reason
4. whether the two separations keep one value at larger sizes is open

USES Directed hypergraph constant 1.14 · first bounds A.67 · bounded outgoing count A.68



$$n = 4, r = 3, m = 3 \quad 8$$

EXTREMAL

one tail of the four, each doing the same

Directed multihypergraph · arc

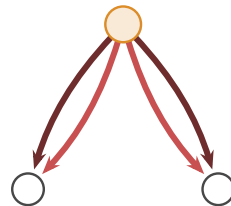
$$L_m^{\text{dir}(r)}(n)$$

UPPER exhaustive enumeration **PROVED**

LOWER $n(m-1)$ **PROVED**

1. the outgoing cut caps each tail at $m-1$ copies, whatever those copies are, so with repeats allowed the count is simply $n(m-1)$
2. at this size a tail has only one possible head set, the other two vertices, so repetition is the only way to grow at all
3. lifting that limit on distinct head sets is the whole difference from the simple model
4. forward, backward and general orientations share the growth rate, backward being the reverse of forward

USES Directed hypergraph constant 1.14 · first bounds A.67 · bounded outgoing count A.68 · backward reverses forward A.69



$$n = 3, r = 3, m = 3 \quad \mathbf{6}$$

EXTREMAL

one tail of the three, each doing the same

Directed multihypergraph · vertex

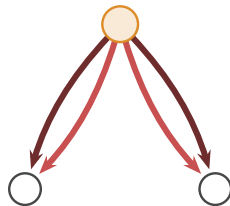
$$K_m^{\text{dir}(r)}(n)$$

UPPER exhaustive enumeration **PROVED**

LOWER $n(m-1)$ **PROVED**

1. the outgoing cut ignores interior vertices, so it gives both separations at once here too and the arc witness carries over
2. the general model, which allows every non-empty tail-head split, obeys the forward bound and shares its constant
3. whether forward and general agree at a fixed $n > r$, and not merely in the growth rate, is open

USES General orientation constant A.72 · general obeys forward A.70 · backward reverses forward A.69 · bounded outgoing count A.68



$$n = 3, r = 3, m = 3 \quad \mathbf{6}$$

EXTREMAL

one tail of the three, each doing the same